Team Boundary Issues Across Multiple Global Firms

J. ALBERTO ESPINOSA, JONATHON N. CUMMINGS, JEANNE M. WILSON, AND BRANDI M. PEARCE

J. ALBERTO ESPINOSA is an Assistant Professor of Information Technology at the Kogod School of Business at American University in Washington, DC. He received his Ph.D. in Information Systems from the Graduate School of Industrial Administration at Carnegie-Mellon University. His research interests include information technology support for geographically distributed collaboration. His current research focus is on global software teams, team cognition, team coordination, and related human factors.

JONATHON N. CUMMINGS is an Assistant Professor of Management at the Sloan School of Management, Massachusetts Institute of Technology. He completed his doctoral dissertation on "Work Groups and Knowledge Sharing in a Global Organization" at Carnegie-Mellon University, where he was a postdoctoral fellow at the Human-Computer Interaction Institute. His research focuses on the social impact of Internet use, online relationships, and electronic communities.

JEANNE M. WILSON is an Assistant Professor of Organizational Behavior at the School of Business Administration at the College of William & Mary. She received her Ph.D. in Organizational Behavior/Theory at Carnegie-Mellon University. Her research focuses on new organizational forms, particularly groups that span geographic and organizational boundaries.

BRANDI M. PEARCE is a Ph.D. candidate in Organizational Behavior at Carnegie-Mellon University. Her research focuses on interorganizational arrangements, groups, and conflict. Her current work examines the dimensions and processes in inter-organizational teams that distinguish them from their traditional counterparts. Her research also includes the examination of context on conflict management and group performance.

ABSTRACT: Numerous methodological issues arise when studying teams that span multiple boundaries. The main purpose of this paper is to raise awareness about the challenges of conducting field research on teams in global firms. Based on field research across multiple firms (software development, product development, financial services, and high technology), we outline five types of boundaries that we encountered in our field research (geographical, functional, temporal, identity, and organizational) and discuss methodological issues in distinguishing the effects of one boundary where multiple boundaries exist. We suggest that it is important to: (1) appropriately measure the boundary of interest to the study, (2) assess and control for other influential boundaries within and across teams, and (3) distinguish the effects of each boundary on each team outcome of interest. Only through careful attention to methodology can

Journal of Management Information Systems / Spring 2003, Vol. 19, No. 4, pp. 159–192.

0742–1222 / 2003 $9.50 + 0.00.

we properly assess the effects of team boundaries and appreciate their research and practical implications for designing and using information systems to support collaborative work.

KEY WORDS AND PHRASES: cross-functional teams, distributed teams, global teams, interorganizational teams multiple boundaries, organizational forms, research methods, virtual teams, work groups.

RECENT TRENDS IN GLOBALIZATION and telecommunications are making new forms of organizing more possible and prevalent [38, 39]. Communication networks and information technologies have provided tools and incentives for linking individuals that are separated by time, distance, or other dividing characteristics [16, 18]. Work in organizations is increasingly carried out across such boundaries, thus creating new work group configurations. For example, Orlikowski's recent study of a global software development organization found that engineers needed to traverse multiple boundaries to do their jobs. Accounts from 78 interviews uncovered the presence of as many as 15 global locations, spanning 19 time zones and representing various levels and areas of technical expertise [41]. The existence of so many boundaries presents a substantial challenge when doing field research on teams. On one hand, it is difficult to identify a fixed boundary around a team that defines the unit of analysis under study. On the other hand, the presence of certain boundaries within the team makes it difficult to tease out the effects of other variables of interest.

Boundaries are the "often imaginary lines that mark the edge or limit of something" (Cambridge Dictionaries On-Line). Team boundaries can be thought of as both the lines within, and the lines around a team—the internal and external "edges" of the team. Chudoba and colleagues described these boundaries in terms of "discontinuities," which they defined as incoherence or gaps in aspects of the team's work, such as the work setting, task, and relations with others [11]. Their literature review of 75 published articles found that studies focused on one or two boundaries, but often ignored other important boundaries [52]. We incorporate this notion of discontinuities in our use of the term "boundary" for the purposes of this study; meaning a discontinuity, edge, or other dividing characteristic present in the work context of a team. Teams have external boundaries that distinguish them from other teams and, as Orlikowski [41] pointed out, they also have internal boundaries (e.g., geography, time zones, functional expertise) that represent edges that team members need to bridge to do their work.

Although research is increasingly focused on new forms of organizing, such as virtual and global teams, very little attention has been given to the methodological challenges of doing field research on teams with multiple boundaries, and we still do not completely understand the implications that arise when work needs to be done across boundaries [32, 49]. As our understanding of these teams advances, we also realize that these boundaries are not always independent. When boundaries are not clear or when they covary, it is difficult to draw firm conclusions about the observed effect of a particular variable or dimension. For example, groups that represent multiple organizations often work from dispersed locations and, although researchers may attempt to draw conclusions about the effect of one type of boundary, such as distance, we often cannot be sure that the effects we observe are not due to other boundaries.

Our review of recent field studies on virtual teams suggests the need for researchers to identify and account for team boundaries in their studies. In the early stages of a study, it is not always clear which boundaries are present [33], thus making it difficult to define the research design and correctly interpret the results. In addition, much of the existing empirical research has identified interesting interactions among boundary variables themselves [4, 13, 25, 34, 35, 45, 53], which highlights the importance of comprehensive consideration and measurement of boundary issues when conducting research involving teams with multiple boundaries.

One of the difficulties of studying team boundaries during field studies is that organizational teams often have numerous boundaries, including geographic, temporal, functional, identity-based, organizational, expertise-related (i.e., novice and expert team members), cultural (i.e., multiple nationalities), historical (e.g., different versions of the same product), social, and political [9, 41, 52]. Consequently, it is impractical to address all boundary issues in a single study. Therefore, consistent with other studies [41, 52], we have narrowed the focus of our study to five specific boundaries that were more prevalent and salient in the four individual field studies on which this study is based [14, 20, 42, 54]: geographical, functional, temporal, identity, and organizational. Two of these boundaries—geographic and temporal—are considered most important when studying teams mediated by technology [7], and are most often found in empirical studies, whereas the other three—organizational, functional and identity—are often neglected [52].

In the next section, we describe the methodology used in this study and the individual research cases on which the study is based. We then discuss the methodological issues that we encountered in each boundary, along with their respective implications for research. We then conclude with a discussion of common observations and overarching research issues identified across all boundaries, and the limitations and final conclusions of the study.

Methodology and Research Cases

THE PRIMARY GOAL OF THIS PAPER IS TO HIGHLIGHT the methodological challenges in studying teams in global firms. To accomplish this, we utilized data from our four separate research studies and conducted a cross-case analysis using a case-oriented strategy [36, 57]. The analysis entailed a two-step process. Step one involved first-level coding in which each case was descriptively analyzed to summarize the methodological issues. During this phase of analysis, the concept of boundaries emerged, providing a framework for subsequent examination. The second step of the analysis involved a pattern-matching technique in which an initial case was examined for methodological concerns associated with each boundary identified and then successive cases were examined to determine whether the same patterns could be found. The results from our two-step analysis are reported in a summary table at the end of the discussion of each boundary.

We will briefly describe the design, context, and data collection methods used in each of our individual studies. The first case, which we will refer to as the "software teams" study, was conducted in a high technology Fortune 500 firm at sites in Germany and England [20]. The data were collected using semi-structured, face-to-face interviews with 36 software developers, technical managers, and project managers, all of whom were working on a particular version of a major software subsystem. All software developers and managers who were available at the time of each site visit were interviewed. All of the German software developers on the team were fluent in English and all the team members were located in the same time zone, though not necessarily the same country.

The second company studied was a Fortune 500 telecommunications firm. Because observations were drawn from groups that worked in product development, we refer to this sample as "product teams" [14]. Insight about the product teams was integrated from three sources: division managers, team leaders, and team members. Division managers provided archival data on the groups, including project documentation, presentation slides, and other available written materials. Team leaders provided group membership information, including names, locations, and functions of the members. Finally, interviews were conducted with the team members and leaders of 20 work teams to learn about issues encountered during their projects.

The third study focused on a financial institution that used teams to manage corporate employee retirement accounts. We refer to this research as the study about "banking teams" [54]. Seventy-eight banking teams were observed and surveyed six times over the course of the teams' first year of working together. In addition to the multiple survey research design described above, three teams were selected for more intensive observation in order to help explain patterns of results from the larger sample of teams. In choosing the three focus teams, the researcher attempted to identify prototypical colocated, distributed, and mixed teams. The focus teams were close to the median values of their respective samples on number of members, number of clients serviced, size of accounts, distance between members, and frequency of face-to-face meetings. Each team member was observed for four hours in months one, two, and three, resulting in approximately 12 hours of observation per member, or about 96 hours per team. All team members in the United States and Canada were interviewed face-to-face, and the two team members in London and one team member in Hong Kong were interviewed over the phone.

The fourth case is extracted from a study designed to examine group processes in teams that cross organizational boundaries and is referred to as the study about "alliance teams" [42]. The data for this study were derived from three sources: team leaders, team members, and archival data such as project documentation, mission statements, rules of engagement, and presentations. The data from team leaders and team members included 14 semi-structured interviews of members from two different interorganizational teams. The first team involved two high-tech organizations working to develop a competency center for technical and marketing outputs. The second team involved two primary organizations (consulting and retail) and three peripheral organizations (manufacturing, high technology, and marketing) working to develop a virtual storefront for selling personalized consumer products.

Geographic Boundaries

GEOGRAPHIC BOUNDARIES ARE PRESENT IN A TEAM when some of its members are separated by distance. A number of research studies have examined the effects of distance on teamwork [1, 12, 50]. However, the effect of geographic dispersion often is difficult to isolate in the face of other factors associated with distant collaborators, such as differing cultures, time zones, and organizations. For example, studies on distributed software teams generally associate geographic separation with increased coordination overhead, more substantial delays, failure to communicate and retain contextual information, and differences in feedback cycles [9, 22]. Armstrong and Cole [4] also found that geographic distance led to misunderstandings and conflict escalation, but they found that other differences between the locations and organizations also contributed to these problems. Another study involving distributed software teams acknowledged that distance alone may not be the only cause of observed problems, but that other factors—the technologies used to communicate, cultural differences, different time zones, and organizational factors—also can play a role [13].

In an article summarizing their findings from ten years of research investigating the effects of distance, Olson and Olson [40] concluded that geographic distance has a negative effect on the development of common ground in communication, but also suggested that some of the problems of geographic dispersion have to do with cultural and functional differences when work is done across sites. Another review of the literature on proximity identified other consequences of distance, such as reduced informal or spontaneous communication, and proposed that the effectiveness of technological and management solutions depended not only on distance, but also on the level of social cohesion in the group [27]. In sum, geographic dispersion has received significant attention in the literature, but much of the empirical work often collapses distance measures with other boundaries such as culture and functions [35]. In this section we discuss methodological issues and solutions when studying teams that span geographic boundaries, as summarized in Table 1.

Software Teams

More than half of the software team members interviewed mentioned explicitly that they did not have many coordination problems with colocated teammates, whereas 91 percent of them reported problems working with colleagues across geographic

Table 1. Summary of Methodological Issues: Geographic Boundary

Boundary	Methodological issues	Potential remedies	Sites where issue was encountered
Geographic (i.e., some members are separated by geographic distance)	Geographic distance confounded with other variables: familiarity, media, expertise, tenure, organizational boundary	Reduce variation on covariates	Software teams, product teams, banking teams, alliance teams
	Geographic distance not exogenous	Sample groups where distance is not self-selected	Software teams, product teams
	Distance is not always a static phenomenon	Assess distance over time	Product teams
	"Meaning" of distance	Theoretically consider the meaning of distance and select appropriate measures	Banking teams
	Nonlinear nature of distance	Triangulate on the measurement of distance	Product teams, banking teams
	Confounding potential of "external" distances	Measure both internal and "external" distances	Alliance teams

locations. However, many of the problems cited were not directly associated with distance per se. For example, many members indicated that a lack of familiarity with colleagues and contexts in other sites (e.g., knowing whom and when to call) were the main problems with distributed collaboration and 69 percent of the respondents indicated that having prior knowledge of colleagues and contexts across sites offset many of the problems with distance. Also, some software team members possessed different levels of tenure and varying types of functional expertise, which made it difficult to maintain smooth working relationships. For example, a larger proportion of German team members had programming expertise and long tenure with the company, whereas a larger proportion of the English group had testing expertise and had only been with the company a few years, with a higher employee turnover. These differences made it difficult to isolate the effects of distance.

When faced with correlates of distance, one way to address the problem is to select the sample in a way that reduces variation among covariates. In this study, we reduced the variation on time zones and familiarity by focusing on developers in England and Germany who were working on a particular software release in which team members had a similar distribution of functional expertise across sites. We limited the study to these two countries because the developers in these two sites often visited the other site, had a history of close collaboration, and had strong similarities (e.g., close time zones, familiarity with other site, fluency in English, same continent, etc.), thus reducing the confounding effects of a lack of familiarity with colleagues and contexts.

The second methodological issue uncovered in the software teams study was that the distribution of work across locations was not always an exogenous factor. A prior study with similar software groups found that their work was often divided and distributed in a way that reduced coordination problems, and that this work tended to be distributed by function, product structure, software process phase, or proximity to clients [21]. Another study with software teams found that geographic distance led to more coordination problems and delays [22]. However, a follow-up study found that when other control variables such as team size and software complexity were added to the model, the direct effect of distance disappeared [23] because more complex projects requiring more diverse expertise tended to be larger and more dispersed across locations. These projects included redundant roles across sites (e.g., liaison engineers, cross-site coordinators, counterparts), thus creating more coordination problems. This issue was addressed in the software teams study by focusing on software developers working on a single subsystem who had a similar mix of functional, product, and software process responsibilities across locations.

Product Teams

Although researchers often refer to a team's geographic distribution in terms of the number of locations where members work, distance is not always a static phenomenon. Group members may travel frequently or change locations because of new buildings, organizational restructuring, or project demands. Therefore, team members are not always in their assigned offices or may move to an entirely different location during a project. This means that static measures of geographic distribution do not always capture the complexity of teams that are distributed in space and time.

Two product teams exemplify how geographic distribution can be more complex than distance alone. First, an electrical circuitry team had nine members from the engineering and quality assurance divisions, who were spread across France and Switzerland. One of the members in this team moved twice during the project, first from Switzerland to France to be with the team toward the middle of the project, and then back to Switzerland toward the end of the project. In a second example, a handheld device design team had ten members from engineering, project management, and customer service, who were spread across locations, including Arizona, Georgia, Illinois, Tennessee, and Israel. One of the members in this team worked at home in Georgia during the project and commuted to Tennessee whenever the team reached key decision points.

Several ways were considered to account for the complexity of geographic distance in these examples. One solution was to divide the project into two stages, early and late, and then measure the extent to which members were distributed at each stage, as well as the change in distribution over time. Alternatively, the distribution could have measured once using the location where each member spent most of his or her time, but since communication was significantly greater for dispersed teams in the later stage than in the earlier stage, it is probably best to know where members are located at each stage. No matter how researchers account for the different patterns of distribution throughout a project, it is crucial to understand whether and how these factors change over a project cycle, and to consider those changes in designing the study methodology.

Banking Teams

In the past, research has tended to show that team members have more synchronous and frequent interaction with their local peers and are likely to identify with local work groups more strongly than with the distributed team. However, as mentioned above for the software teams, location may not be an exogenous variable. Organizations make decisions about where employees will be located for all kinds of reasons, which may affect dependent variables that researchers are interested in studying. For instance, in previous experiments with distributed work at the bank we studied, team dysfunction caused managers to consciously decide to relocate some members back to the main headquarters in an effort to create more cohesive teams. Because of this, any attempts to draw conclusions about the effects of colocation would have been clouded by an unspecified causal relationship between team performance and the independent variable of interest (geographic dispersion).

Once other related variables have been identified, researchers are still left with the problem of how to effectively measure distance or geographic dispersion in teams. For instance, in one team at the bank, one member was located on the 11th floor of a downtown office building, another member was located on the 9th floor of an adjacent office building connected by a sky bridge, the third and fourth members were located in a back-office operation about five miles from downtown, and a fifth member was located approximately 757 miles away in Atlanta. In this case, most of the analysis was conducted at the dyad level. Each member of the team rated how much they interacted with and trusted every other member of the team. The distance between each pair of members was easily calculated and used in the analysis. However, it is more common in these types of studies to conduct the analysis at the group level. A group-level measure of geographic dispersion based on the greatest distance between team members (a measure that has been used in the past), would not adequately reflect the clustering of most team members in one city. It may be more appropriate to use the weighted average of travel time between sites. Alternatively, the location of some members may be more important than others. Consider the difference in meaning if the team leader was located in Atlanta rather than the headquarters city. In that case, average distance from the team leader might be a more appropriate measure of distance.

Alliance Teams

Measuring only the distance between members within a team may not fully capture the impact of geographic dispersion in teams that cross organizational boundaries. In the case of the alliance teams drawn from the research sample, teams were configured in four main ways: (1) members of the team were colocated at a site that was physically separate from all participating organizations; (2) members of the team were colocated with their respective home organizations and worked in a distributed fashion with members of the team; (3) members of the team were colocated at one of the participating organizations, thus some members worked remotely from their home organization; or (4) some members were colocated whereas other members were distributed. From these examples, it becomes evident that distance in an interorganizational context has two dimensions: internal distance (distance between members) and external distance (distance between members and their home organization).

When assessing geographic distance, researchers need to consider the relevance of external distance. One way to address this is to measure the physical distance between members while also measuring the distance between members and their home organizations or another relevant external constituent. Based on the purpose of a particular study, the data may be used as part of the analysis or as a control. Another option is to reduce the variability of external distances by sampling on one of the specific configurations listed in the above section, rather than drawing data from all four.

Functional Boundaries

FUNCTIONAL BOUNDARIES ARE PRESENT when more than one area of functional expertise is represented on a team, such as marketing, engineering, and manufacturing

[2, 15, 19]. Because functions often are interrelated with one's knowledge, skills, and experience, measures of cross-functionality in teams are not necessarily independent from other important measures. In addition, when each function is housed in a separate location, function and distance measures may be confounded, making it difficult to assess whether particular work patterns are a result of differences in functions or in distance, or some interaction of the two. Other research indicates that functional diversity can influence other team processes, such as task conflict, which in turn can affect the group's performance [26, 43].

In order to avoid these pitfalls, researchers must consider how functional boundaries influence the processes of a team and whether they might interact with other existing boundaries. In this section, we discuss methodological issues and solutions when studying teams that span functional boundaries, which are summarized in Table 2.

Software Teams

The study initially focused on feature development teams, which were responsible for adding incremental functionality (i.e., features) to a working product. It became clear, however, that the feature development team concept was new to the groups under study. They viewed the implementation of a feature more as a task than as teamwork. Furthermore, the feature development teams were strongly divided by functional boundaries, and most interviewees had very strong bonds to their functional groups. These strong ties made it difficult to study the effects of other boundaries that were present, because strong functional boundaries can change the meaning of those other boundaries. We observed this effect when comparing a group of testing engineers in England and Germany whose shared familiarity with technical terminology, procedures, work schedules, and deadlines helped them collaborate more effectively and share more common ground—despite their geographic distance—than a group of technical specialists located in the same office building in Germany who had more diverse functional expertise. The latter group appeared to collaborate less effectively with each other as a result of their specialized knowledge and terminology. This issue was addressed by selecting an even mix of functional and software process specialists from both geographic locations in order to account for the functional differences by location.

Product Teams

An important observation that arose in the study of cross-functional product teams was the distinction between *perceived* and *actual* functionality. Team leaders were asked in a survey to indicate the functional area of each team member. The leaders indicated their perceived functionality of the team rather than the functions provided by the company database records. However, the responses to this question were not uniform across teams. Leaders tended to report greater functional similarity within a team when the disciplines were distinct (e.g., engineering, manufacturing) compared to teams where the disciplinary boundaries were less clear (e.g., design engineering

Table 2. Summary of Methodological Issues: Functional Boundary

Boundary	Methodological issues	Potential remedies	Sites where issue was encountered
Functional (i.e., more than one area of functional expertise is represented in the team)	Functional affiliations may change the meaning of other boundaries	Sample groups with an even mix of functions across locations	Software teams
	Differences between perceived and actual functional representation	Use both indices: self-report and objective measures	Product teams
	Function confounded with other variables: expectations, reputation, power, authority	Measure perceptions of expectations, reputation, power, and authority of function	Product teams, banking teams, alliance teams
	Function confounded with other boundaries: geographic distance and organizational boundaries	Sample teams with similar functions or similar distributions of functions across distance or organization	Software teams, product teams, banking teams, alliance teams
	Function interacts with other boundaries	Sample groups with similar functions or measure differences in function and control analytically	Product teams, banking teams, alliance teams

and layout engineering rather than "engineering"). The more similar team members were, the more leaders attempted to differentiate their functions.

Two of the product teams illustrate the contrast between perceived and actual functionality. First, a circuit design team had nine members from engineering, who were spread across the United States, Israel, and Singapore. Their task was to design chips that could process digital signals more efficiently in electronic products. Even though the nine members in this team represented the company's engineering function, the leader reported that five of these members were in "design" engineering, whereas the other four "layout," "product," "field application," and "computer-aided design" engineering, respectively. In the second instance, a mobile manufacturing team had 11 members from engineering and manufacturing, who were spread across the United States, Germany, and Ireland. Instead of specifying the types of engineers, the leader of this team reported that eight were from engineering and three were from manufacturing.

When assessing cross-functionality, researchers need to be clear whether they want to measure perceptions of functional differences among team members (as team leader surveys would yield) or actual differences among team members (as company records would show). One solution used here was to create two indices of cross-functionality, one in which team leader or member distinctions were used, and another in which the company distinctions were used. If company records are not available, alternative coding schemes can be devised to differentiate between true functional differences, taking into account potential bias on the part of team leaders or team members' attempts to distinguish themselves from others.

Banking Teams

Although functional roles are seemingly easy to measure, function is often confounded with other important variables of interest. This was certainly the case in the banking teams; but the related variables were not obvious or immediately apparent. Several months of on-site observation of three representative teams revealed that function was sometimes confounded with geographic distance as well as with status and preconceived expectations about team members' behavior. For instance, in the case of the banking teams, one whole function (the accounting or information delivery analysts) was located in a "backroom" technical center several miles from the downtown headquarters of the bank, whereas most (but not all) of the relationship managers were located in the headquarters building. This meant that at least these functions covaried largely with geographic distance. In this setting, it would have been impossible to determine whether low trust between the relationship managers and the accountants on each team was due to distance or due to functional differences. This situation is often true in organizational settings where the R&D facility is physically removed from the main office, and customer service or other "back-office" functions are relegated to other locations. In the example given, it was possible to differentiate the effects of geography and function because other functions were less systematically distributed.

Another issue that surfaced in early observations and interviews was that team members from some functional areas had reputations that affected team performance and behavior. For example, some team members tended to view the risk and performance analysts as picky and detail-oriented. Other members expected these analysts to examine the team's monthly reports and publicly blame others for any errors they found. These expectations resulted in a form of confirmation bias; for example, the accountants did not check the foreign currency rates for the client's global investments very carefully because they felt that they could count on the risk analysts to catch mistakes. This cycle of behavior—in addition to causing plenty of mistakes—ultimately deteriorated the relationships between the accountants and the other analysts.

There are numerous methodological concerns with analyzing the situation described in the above paragraph. If a measure of each team member's function was being used as a proxy for "unique knowledge contribution," as it often is, there would be no relationship between diversity of function and knowledge sharing in this case. The lack of communication and disregard for the other function's expertise—which covaried with function in this case—would act as suppressor variables, producing spurious results in the analysis of the relationship between function and the dependent variable of interest. One solution to the problem is to measure the effects of role reputation before the start of the teams and to use this as a control variable in the analysis.

Alliance Teams

Interorganizational arrangements are often designed to leverage the core competencies of the participating organizations. Thus, it is common to have interorganizational teams where functional boundaries covary with organizational boundaries. This makes it difficult to assess whether outcomes are associated with functional differences or organizational differences. For example, in one alliance team used in this study, all team members from organization A were engineers, whereas all members from organization B were marketing specialists. One mechanism for resolving this issue is to sample teams where members represent the same functional backgrounds or to sample teams where there are similar distributions of functions from both organizations.

When teams are embedded within one firm, members share a context that is internal to the organization although it may be external to the group. Concomitantly, even though multiple functional areas may be represented on a particular team, its members will tend to have similar understandings of the authority and power associated with particular roles. In contrast, the members of teams that cross firm boundaries may have very different perspectives regarding the authority and power associated with a particular function. Interviews from an alliance team used in this study revealed that in organization A, engineering was perceived as a powerful role. However, in organization B marketing was seen as an influential role and engineering was perceived as a back-office function with limited authority and power. As a result, it becomes difficult to determine whether specific group processes, such as conflict, are linked to functional differences or to different perceptions of power and authority. As in the case of the banking teams, it may not be enough to measure the different types of functions represented in a team. Researchers may also need to measure how roles are linked to perceptions of authority and power so that these factors can be controlled for.

Temporal Boundaries

TEMPORAL BOUNDARIES ARE PRESENT IN A TEAM when some of its members are separated by time because of differences in working hours, time zones, or working rhythms that reduce the time available for same-time (i.e., synchronous) interaction. Global teams operate in different geographical locations, but they often operate in different time zones as well. When studying the effects of various factors on team outcomes, it is important to distinguish whether teams are separated by time or distance [7]. For example, teams separated on an east–west axis have fewer overlapping work hours than teams separated from north to south [39], thus making it more difficult for the former to coordinate and communicate. Temporal boundaries can also exist within teams because of differences in working hours, as in the case of multiple shift work [3].

When teams are separated by distance and their communication is mediated by information technologies, temporal boundaries may exist regardless of time zones and working hours, depending on the frequency and synchrony of the teams' interaction patterns. Some teams may communicate frequently using synchronous technologies such as telephone and videoconferencing equipment. Other teams may interact asynchronously using different-time technologies like electronic mail and shared databases. Some have argued that computer mediation presents a unique context for interpersonal communication and it has its own temporal qualities [24]. Further, members of global teams often face different temporal pressures coming from varied corporate schedules and priorities, delayed feedback cycles, and different communication technologies. These pressures challenge teams to manage the timing of their activities and challenge researchers to accurately assess the consequences of temporal boundaries, particularly when they can be confounded with the effect of other boundaries. In this section, we discuss methodological issues and solutions when studying teams that span temporal boundaries, summarized in Table 3.

Software Teams

Large-scale software development work is inherently asynchronous. It is characterized by a large number of individuals and teams working simultaneously on different parts of the same software product, which requires a substantial amount of temporal coordination. Seventy-five percent of the respondents identified temporal coordination problems that surfaced because time dependencies among individuals and teams were not effectively managed. Examples of these problems include software parts and activities that were not finished on time and unsynchronized project schedules.

Table 3. Summary of Methodological Issues: Temporal Boundary

Boundary	Methodological issues	Potential remedies	Sites where issue was encountered
Temporal (i.e., some members are separated by time—differences in working hours or time zones)	The asynchronous nature of a task may be moderated by geographic distance	Sample on teams with similar temporal boundaries and time zone differences	Software teams, product teams, banking teams
	Geographic distance and temporal boundaries are more likely to covary more so when different time zones are involved	Measure and control for temporal boundaries or sample on teams with similar temporal boundaries	Software teams, product teams, banking teams
	Task interdependence and communication media are often inherently tied to temporal boundaries	Measure other variables caused by temporal boundaries and control based on a theoretical consideration of their relationship to temporal boundaries	Product teams, banking teams
	Temporal boundaries can be confounded with national-cultural boundaries	Qualitatively analyze the causes of group behavior. Consider measuring national and cultural affiliations as potential control variables	Product teams, banking teams
	Patterning of temporal boundaries	Utilize dispersion or network measures of density and centrality	Banking teams
	Temporal boundaries covary with organizational boundaries	Measure perceptions of work tempos, size of participating organizations, and preferred modes of communication	Alliance teams
	Confounding of "external" differences in time zones	Measure differences in "external" time zone as well as observing the effects	Alliance teams

The asynchronous nature of software development makes it difficult to isolate the effects of distance or other boundaries from temporal effects; one may conclude that coordination problems are more severe in geographically distributed work than in colocated work when some coordination problems can actually be attributed to different work schedules and time zones. For example, the effect of geographic distance on coordination is less likely to be confounded with the effects of temporal boundaries for teams with members working in England and Germany than for teams with members working in England and the United States because the latter two countries have less overlap in their working hours. This problem was addressed in the software teams study by interviewing developers only in England and Germany.

Product Teams

Without specific knowledge of the task dependencies and communication media required to accomplish a task, interpreting the impact of temporal boundaries on performance can be difficult for groups with members spread across time zones. For example, if the work is very interdependent and members need synchronous communication to do their jobs effectively, then time zone differences can potentially disrupt coordination because they force members to communicate primarily via asynchronous technologies. Not being able to pick up the phone and call other members because they are sleeping or out of the office can really slow down a group's progress. However, if the work is not very interdependent and can be done with asynchronous media, then time zones could actually enhance the pace of work—members working in one part of the world could "hand off" their work at the end of the day to be continued by members in another part of the world. At a minimum, members could work on their pieces of the project until they are complete, without concern that another piece of the project is waiting.

A comparison of two highly interdependent product teams highlights the difficulty identifying whether time zone differences influence performance. The first team had nine members in Florida (6), Georgia (2), and Denmark (1) in which the member in Denmark was 6,300 miles and six time zones from the other members. The second team had 11 members in Ireland (10) and Germany (1) in which the member in Germany was 3,500 miles and one time zone from the other members. Even though both distant members where sufficiently far away from the others to restrict face-to-face communication, the member in Denmark relied almost exclusively on e-mail to communicate with the U.S. members, whereas the member in Germany was able to regularly use the telephone to communicate with the members in Ireland. In the eyes of senior management, the second team performed better than the first team, in part because of their teamwork and ability to share knowledge effectively. What is difficult to know, however, is whether the large time zone difference or their reliance on asynchronous media (or both) disrupted their teamwork and knowledge sharing.

When focusing on the effect that temporal boundaries have on team outcomes like performance, it is critical to measure other variables influenced by temporal boundaries (e.g., use of e-mail versus phone calls as a communication medium) that could also affect performance, making it is difficult to isolate the related effects. Researchers need to be aware of the unintended consequences of variables like time zone differences, which often blend distance, culture, and media together.

Banking Teams

Although many of the teams in the banking case were geographically distributed, only a few spanned more than one or two time zones. In this case, spanning multiple time zones often coincided with the presence of other boundaries and a marked decrease in face-to-face meetings of the full team. Because there was some variance in distance–time zone relationships, it was possible to qualitatively examine the separate effects of distance and time. For instance, one team had a member in San Juan and another member in Montreal (a distance of 1,920 miles, but in the same time zone). Such cases could be compared to dyadic relationships that spanned New York and London—about the same distance—but time zones apart.

Qualitative observations suggested that distance affected teams' methods of communication and the salience of remote team members, whereas spanning multiple time zones affected the rhythm of the team's work and created unexpected faultlines in the group [31]. In one case where most of the U.S. members of a team worked in the Central time zone and some members worked in London, the team had to organize its work so that the critical pieces of information needed from London were obtained by 11 A.M. Central time. However, the London-based members were unavailable to deal with any problems that cropped up in the afternoon. The U.S. team members had difficulty trusting that problems they encountered could be left for the London members to resolve the next morning, because many of the problems had to be resolved by particular deadlines and it was not always possible to assess how complex the problem would prove to be. This led to a pattern of behavior in which the U.S. subgroup did not think to include the London members in problem-solving activities, even when they could have been helpful. Naturally, the London subgroup often felt excluded or marginalized. Without qualitative observation, the conflicts in these groups may have been erroneously attributed to national or cultural differences, when in fact they were caused by patterns of problem-solving behavior that stemmed from time zone differences.

As we noted in the discussion of geographic boundaries, the patterning of temporal boundaries may be more important than the simple range of time zones spanned. In other words, having many team members clustered in one time zone may have a greater effect on team dynamics than having most members spread out across multiple time zones. This possibility suggests that researchers should consider assessing different forms of functional relations by including dispersion measures [10] or even network measures of density and centrality [51] to adequately represent the complex patterns of dispersion within global teams.

Alliance Teams

Organizational practices affect preferences regarding communication media (phone, e-mail, meetings, etc.), the rhythm of work, and the timing of interactions. To be successful, interorganizational teams have to integrate their organizations' multiple temporal contexts. These considerations are in addition to—and may complement or conflict with—the temporal context within the group.

In an alliance team observed in this study, organization A preferred using e-mail as a means of communication and process flow, whereas organization B preferred face-to-face meetings or telephone calls. Organization A had fewer hierarchical layers and more bidirectional communication, whereas organization B was nearly ten times larger with more layers of bureaucracy, requiring more time to acquire resources and necessary approvals. What on the surface appeared to be a by-product of an organizational boundary was actually the result of the different temporal contexts of each organization. Thus, it is important to assess temporal factors that may covary with different organizational contexts such as the perceptions of the participating organizations' work tempos, the size of the participating organizations, and the preferred modes of communication.

In addition, members of an interorganizational team may be located in different time zones than their parent organizations. Because the internal dynamics of such teams are heavily influenced by members' interactions with their home organizations, it is important to not only observe the effect of different time zones internally among members, but to also consider the effects of time zone differences between members and their home organizations or external constituents.

Identity Boundaries

IDENTITY BOUNDARIES ARE PRESENT when some members of a team are not fully dedicated to the team, either because they are working on multiple projects with multiple teams or because their teams are nested within larger teams. Team members often are involved in more than one project, team, or organizational unit at a time, making it difficult for them to define their many identities. Members may also have started a new project before finishing the current one, or the project they are working on may be part of a larger project that has complex interdependencies.

When team members are part of multiple projects or teams, the challenge for researchers is to distinguish the impact of participation in one group from the impact of participation in the remaining groups. This is because identification can have powerful effects on behavior (e.g., satisfaction, turnover, and performance) [5]. Identity boundaries may also interact with other boundaries in complex ways. For example, one survey study of 276 employees from the sales division of a large international computer company that had a mandatory telecommuting program found that higher levels of electronic communication were associated with increased organizational identification for employees who worked few days in the office. However, higher levels of phone communication were associated with *decreased* organizational identification for those same employees, whereas it was associated with *increased* organizational identification for employees who worked many days in the office [53]. Since increasing team or organization identity has been suggested as a critical mechanism in holding virtual organizations together [37], and because identity often interacts with other boundary conditions, identity is an important consideration in any study of global teams. In this section, we discuss methodological issues and solutions when studying teams that span identity boundaries, which we summarize in Table 4.

Software Teams

Teams sometimes are nested within larger teams. In the software teams study, software modification teams were nested within feature teams, which were in turn nested within larger product release teams. This made it difficult to identify the proper level of analysis. The critical goal in cases like this is to measure variables at a level where the team identity is meaningful. In the software teams, developers often did not identify strongly with their feature teams because developers worked on multiple features of the same product release in parallel. Consequently, team members exhibited a stronger sense of identity with their larger product release team than with the smaller feature team. Product release teams had well-defined external boundaries and their members were almost fully dedicated to the particular product release to which they were assigned. In contrast, other software divisions in this company employed feature teams in which developers were fully dedicated to single features, and these developers identified strongly with their respective feature teams.

Based on the findings discussed above, it was decided to focus initially on the product release team as the unit of analysis. Thus, we interviewed mostly developers and managers from one product release team. In a subsequent survey study, however, we were more interested in investigating underlying work processes and dependencies that exist when programming software, so we focused on smaller software modification teams ranging from one to ten developers. Whereas developers were not always dedicated to a single software modification, they did have a stronger sense of identity with the particular modification in which they were involved at that time. Also, studying only programming teams eliminated functional boundaries that could have otherwise introduced confounding variables. By recognizing the best ways to focus our research, we were able to develop a deeper understanding of the coordination processes at the root of the software development process.

Product Teams

Whereas work group boundaries often are set by who is “in” and who is “out,” there are substantial differences across teams in terms of how much time members commit to the team. In the sample of work groups discussed here, the average time commitment to each project was 60 percent, which meant that 40 percent of a member’s time at work was generally devoted to another project. Furthermore, not all team members within a group devoted the same proportion of time to the project. Some members participated full-time, but others only participated part-time or as needed. Understanding

Table 4. Summary of Methodological Issues: Identity Boundary

Boundary	Methodological issues	Potential remedies	Sites where issue was encountered
Identity (i.e., some members are not fully dedicated to their teams; they also work with other teams)	Multiple membership affects identity/affiliation	Identify the appropriate level of analysis for meaningful team identity: project, event, group, or work process	Software teams, product teams, banking teams, alliance teams
	Multiple membership violates independence assumptions in many statistical tests	Exclude teams with overlapping membership	Product teams, banking teams
	Involvement in multiple projects, events, groups or work processes impacts identification	Control for level of involvement in a group, task, event, or work process	Software teams, product teams, banking teams, alliance teams
	Time commitment differs across members and teams	Control for amount percentage of time as well as variation of time commitment within a team	Product teams, banking teams, alliance teams
	Membership may be dynamic thus making it difficult to define	Include members if the team leader saw an individual as a member of the team	Product teams, alliance teams
	Identity is a powerful influence on behavior and does not always covary with time spent working on team activities	Measure group/unit identity directly. Consider measures of rewards and accountability as proxies for identity	Banking teams, alliance teams
	Identity is dynamic and can shift over time	Measure identity at multiple points in time	Banking teams, alliance teams

how much time and energy each member invests reveals their stake in the group, which often plays a role in the strength of their identification, though time commitment is not always a perfect proxy for group identification.

Two examples provide a good comparison that shows differences in time commitment or identification. First, a material bonding team had ten members from engineering and quality assurance, all of whom worked in Malaysia. The percentage of time these members spent on the group project ranged from 60 to 70 percent, which meant that all members of the team were committed in similar ways. In contrast, a memory chip team had ten members from engineering and project management, all based in Texas. However, the percentage of time these members spent on the project ranged from 40 to 90 percent. Thus, there was much less consistency in the time commitment of the second group.

In terms of defining membership for a team, one solution is to include members if the team leader sees an individual as part of the team. However, going beyond dichotomous decisions about who is in or out, it is also important to control for the extent of membership, such as the time commitment of members. Disparities in participation or identification in the team may be a source of conflict for members, and researchers interested in the individual reactions to group processes need to pay particular attention to time commitment. For example, when assigning responsibility for parts of a task, time commitment can be extremely important because if one member is only spending 30 to 40 percent of his or her time on the task, then it will take longer for this member to produce a piece of work than it would take a member who is giving 80 to 90 percent of their time.

Banking Teams

As is common in many organizations today, some members of the banking teams belonged to multiple teams. For instance, one information delivery analyst tracked the accounts of both General Motors (GM) and Advanced Micro Devices, making her a member of both teams. The obvious issue in situations like this is that if people are members of more than one team included in the analysis, researchers may be violating the independence assumption required in many statistical tests. One answer to this problem is to exclude teams in which any of the members overlap with teams that are already in the sample. Although this solves the problem of independence, it does not address all of the issues associated with identity in these cases.

There are several implications of belonging to more than one team. One is that an employee who is in multiple teams simply cannot devote as much time to each team as someone who belongs to only one team. In the case of the analyst mentioned earlier, her ratings of the team were based on a smaller sample of team behavior than the members who were assigned full-time to only that team. This can be handled by measuring the percent of time that each member spends on team activities, and using time as a covariate in the analyses. A second issue is that employees may feel more or less identified with the different teams, and identification with a group may not necessarily be tied to the amount of time spent with that group. This can be seen in the case of the information delivery analyst, who started her career at the bank working on the GM account. Even though she now spends more of her time on the Advanced Micro Devices team, she continues to think of herself primarily as a GM team member, because GM is a prestigious client. The fact that identity does not always covary with time spent in a group suggests that in groups composed of members with multiple affiliations, group identity may need to be measured directly.

Alliance Teams

Inherently, members of interorganizational teams have multiple affiliations. In the case of the alliance teams, members were affiliated (at a minimum) with their parent organization as well as the alliance team. Most of the members were also involved with other groups within their home organizations. In one particular alliance team, members from organization A spent 80 to 90 percent of their time working as part of the alliance team, whereas members from organization B only spent 20 to 30 percent of their time working as part of the alliance team. This observation reveals the tight coupling of identity boundaries and organizational affiliations. Similarly, members from organization A were rewarded and held accountable for their performance in the alliance team. However, rewards for members from organization B were based on performance only in their home organization. Consequently, members' sense of identity varied significantly depending on their organizational affiliation. Interviews also revealed that members' perceptions regarding their identity changed over time. For example, in the alliance team noted above, members from organization B indicated that in general their sense of identification was linked to their home organization. However, during periods of intense activity on alliance-based projects their identification with the alliance team increased. To address these issues, researchers may need to measure identity over time, consider proxy measures for identity such as rewards and time spent working with the team, and carefully consider the level of analysis.

Organizational Boundaries

ORGANIZATIONAL BOUNDARIES ARE PRESENT in a team when its members belong to more than one organization. An important implicit assumption that has permeated most team research is that members belong to a single organization. As our understanding of new forms of organizing evolves, it is apparent that interorganizational arrangements, such as outsourcing, joint ventures, partnerships, and alliances, are leading to an increase in the utilization of groups that cross organizational boundaries [33, 56]. Whereas these boundaries are not always specified, research on distributed work groups often includes teams that have members from multiple organizations [4, 13, 45]. This can have complex implications when interpreting results. Differences in organizational affiliations can reduce shared understanding of context and can inhibit a group's ability to develop a shared sense of identity. For example, a field study investigated the effect of social context on appropriation of communication technology by contrasting how two news organizations used electronic messaging. The study found that groups that were similar with respect to functional structure, task process, technology, communication mode, and tenure, but had different social contexts with respect to communication climate, management philosophy, and cooperation, had similar patterns of face-to-face and electronic messaging interaction within their group, but very different patterns across groups. This led to different levels of communication and performance effectiveness [58]. Given that organizational boundaries are often unspecified in field research studies, it is important to acknowledge its potential direct effects while also considering how organizational affiliation may covary or interact with other boundaries such as time and identity. In this section, we discuss methodological issues and solutions when studying teams that span organizational boundaries, which we summarize in Table 5.

Software Teams

The software teams interviewed in this study had a fair amount of collaboration with developers in India where two different organizations were involved: a group from within their company and an outsourced group that employed sophisticated software processes and highly skilled software engineers. Surprisingly, study participants expressed a general preference for working with the outsourced group. Whereas both groups were technically competent, the in-house group was more hierarchical and more difficult to communicate with because formal channels had to be followed and it was not easy for developers to talk directly to managers and directors above their own levels in the organizational structure. In contrast, the developers were able to communicate directly with their collaborators in the outsourced group, many of whom traveled to both England and Germany to be trained for extended periods of time, thus building cohesion and stronger bonds with their team members despite their organizational affiliation. On the other hand, the company used secure communications and firewalls, making it much more difficult for the outsourced group to access resources inside the company's secure intranet.

The different communication patterns and access to internal information we observed between the in-house and outsourced groups made it difficult to study the effects of other boundaries (i.e., geographic distance) with the Indian groups because we could not always distinguish the effects of distance from those of organizational practices. We resolved this issue by interviewing developers in England and Germany only, thus excluding the outsourced groups. However, we asked interviewees to describe their interactions with their Indian peers to gain some insights into these working relationships.

Product Teams

Some of the product teams included members from different divisions or members who represented the customer organization. Issues arise for groups whose members

Table 5. Summary of Methodological Issues: Organizational Boundary

Boundary	Methodological issues	Potential remedies	Sites where issue was encountered
Organizational (i.e., members belong to more than one organization)	Organizational practices confounded with other boundaries such as distance	Measure and control for organizational boundaries	Product teams
	Interorganizational groups and internal groups have distinct group processes	Do not use both interorganizational and internal teams in the same sample	Software teams, product teams
	Some teams include members from external organizations	If these teams are to be used in the analysis ensure that all members are included in assessments regardless of organizational affiliation	Product teams, alliance teams
	Members exist in two contexts	Measure variables in the home context	Software teams, product teams, banking teams, alliance teams
	Organizational affiliation does not exert a uniform influence	Observe and measure patterns of behavior	Banking teams, alliance teams
	Variation in number and patterning of organizations represented	Sample to minimize variance or measure and control analytically	Alliance teams
	Organizational boundaries interact with other boundaries	Sample groups that have similarities across other boundaries, or measure and control for other boundaries	Software teams, product teams, banking teams, alliance teams

come from different organizations, because they are less likely to share common ground and experiences and may disagree about how to allocate the group's resources. For example, conflict can arise if one organization offers its members higher pay, better accommodations, or more administrative support than members from other organizations receive.

Two of the product teams provide examples for how group members can span organizational boundaries. One team had five members from engineering, who worked in Scotland, charged with increasing the quality of silicon wafers for transistor manufacturing. In this team, two of the five members were from the supplier organization, which allowed them to supply materials needed for the project. A second team had eight members from engineering and manufacturing, who were spread across Texas and Arizona. In this team, four of the eight members were from a separate company, which was later purchased by the larger company during the project. In this case, members essentially became part of the same company over the project's life span.

To determine member affiliations and the impact of being a part of different organizations, feedback was solicited from all parties involved, not just the members of the parent organization. Interviewing only some of the members could leave out important information about the group, which may in fact yield the most interesting insights. Moreover, the possibly different contexts of members from all organizations should be considered in order to uncover other possible influences on group behavior, such as the availability of resources.

Banking Teams

Although the banking service teams were generally composed of members from within the bank, other teams within the larger financial services organization crossed organizational boundaries. In the cross-organizational teams, members managed their time commitments by either exaggerating or downplaying the pressures they felt from their home organization. In cases where a member wanted to minimize her commitment to the cross-organizational team, for example, she would overstate the size of her workload in the home organization. In other cases, a member who really wanted to participate in the cross-organizational team may fail to report all the pressures he was experiencing in his home unit. The implication for the study of cross-organizational teams is that it is very important to measure not only what is going on within the team, but also pressures and influences coming from the home environment. As in the case of within-organization identity issues, members often are positioned in more than one camp, and ignoring critical parts of their jobs may result in a skewed view of the situation.

Moreover, it is not possible or advisable to assume that organizations exert a uniform influence on members in these cross-organizational teams. Certainly, they do not exert a uniform "pull." In some cases, the influence may be a "push," as in the case of a dissatisfied employee who would like to get a job with one or more of the other organizations involved in the team. Team members' loyalties, behavior, and decisions are not always aligned with those of their home organization.

Alliance Teams

As very little research has focused on organizational boundaries at the group level, there also has been little focus on the methodological implications of organizational boundaries. Observations from the alliance teams case study suggest that strong organizational affiliations interact with other boundaries. For example, in one joint venture team, the geographically dispersed members were affiliated with separate organizations that had different practices and demands. It is possible that the distance between members, in combination with the organizational boundary, intensified their inability to understand one another's context and exacerbated the negative consequences of the organizational and distance boundaries. Thus, it is important for researchers to theoretically determine whether two or more boundaries may interact and to include them in the analysis when appropriate.

In addition, when examining teams that cross organizational boundaries, it is important to consider the number of organizations represented. Preliminary observations from this study suggest that the number of organizations involved in an interorganizational team may affect the processes that occur within the team. In comparing two interorganizational teams, it became apparent that more organizations resulted in less coordination. If variance on this dimension cannot be reduced through sampling, researchers should consider accounting for it by measuring the number of organizations involved and controlling analytically for it. Similarly, it may be important to consider the actual distribution of organizations represented within a team. For example, a team with three members from organization X and three members from organization Y is likely to result in different group processes than a team that has one member from organization X and five members from organization Y. Researchers can minimize this variation by sampling on teams with similar organizational representations or can control for it by measuring the number of members per organization in any given team.

Common Observations

As our understanding of groups that cross boundaries continues to evolve, it is important to recognize the complexity of the underlying context and how it affects both the design and the implications of our research. Teams often have multiple boundaries defined by location, function, time, identity, and organization. As a result, researchers are faced with a number of challenges when studying such teams. We next discuss some of these challenges, in the form of answers to methodological questions we encountered in our studies. It is important to note that although we focused on five boundaries, many other boundaries may be present within teams (e.g., expertise, cultural, political, etc.). Whereas it would be impractical to discuss every possible boundary, researchers need to be aware of these boundaries and how they may affect research results. Some boundaries may be more salient or important for some research designs and team compositions than for others, so it is up to the researcher to identify which boundary effects may covary with other study variables and implement appropriate controls. Based on our cross-case analysis, we have identified the following overarching issues.

What Should Be Included in the Unit of Analysis?

In increasingly fluid organizations, it can be difficult to decide who is a member of the team and who is not. Team membership can be defined broadly to include multiple boundaries (e.g., members from other organizations, members with a small time commitment to the project), and then measure, or control for the presence of multiple boundaries. Alternatively, team membership can be defined more narrowly, thus minimizing the need to measure and control additional boundaries. The former option introduces possible confounds, whereas the latter limits further understanding of cross-boundary effects. Thus, researchers need to understand these trade-offs when designing their studies. In the product teams, for example, team leaders assessed who was a team member and who was not. Whereas some tasks were completed by non-team members, the ultimate responsibility for a product's success or failure lay in the hands of the members. When it is clear that team functioning is influenced by external stakeholders, one solution is to include these stakeholders in the study sample. Similar critical and other unexpected insights have surfaced based on the qualitative study of the contexts in which global virtual teams operate [13, 25].

Which of the Multiple Boundaries Do We Need to Control For?

When multiple boundaries are present within teams, many of them represent potential confounds. Geographic distance, in particular, can be vulnerable to other confounding factors [4, 35]. For example, whereas colocated software teams tend to be more coordinated than geographically distributed ones, the increased coordination could also be due to stronger similarities in functional expertise and identity affiliations within the team. Similarly, if we are studying coordination across functional boundaries, it will make a difference if those functions are colocated or geographically distributed. Consequently, research on global teams with multiple boundaries needs to control for competing membership in other functional, identity, or organizational groups whenever it is anticipated that such boundaries may affect the dependent variable under study.

Which Interactions Do We Need to Explore?

While accounting for the main effects of particular boundaries, it is quite likely that some of these boundaries will interact with one another. For example, some members of the banking teams who were in different geographic locations also represented different functions, so it is possible that the negative consequences of distance (e.g., less frequent communication) were compounded by the negative consequences that can occur when members span different functions (e.g., less common ground in their communication). In other words, whereas it may be difficult for team members to communicate across geographic boundaries, the difficulty could be compounded if they also have to communicate across functional boundaries. It is up to the researcher to determine whether and when certain boundaries interact, and therefore include interaction terms as necessary.

How Should We Measure Boundary Variables?

The particular boundary measures used in any given study may depend on which team characteristics are most important for the research design. Thus, it is important to consider what a given boundary means in the specific research context and then select one or more measures that represent the most relevant aspects of those boundaries.

Geographic Boundaries

When it comes to measuring geographic boundaries within teams, there are a number of possibilities. For example, O'Leary and Cummings [39] have outlined several measures of distance or dispersion: (1) number of sites represented within the team, (2) degree of isolation (measured by dividing one by the average number of team members per site) to address the characteristics of sub-locations within the team, and (3) separation of sites (the weighted average travel time between sites). They also suggest two additional measures that address some of the confounds typically associated with raw distance measures: (4) a role index that reflects the fact that distance from some members (particularly the leader) is relatively more important and (5) an external index that reflects the fact that distance from some outside constituents (such as the customer) is relatively more important than others.

When measuring distance within dyads, however, it is important not to confound the measure of distance with the outcome of interest. For instance, the specific form of the negative relationship between geographic distance on social influence was called into question because the measure of social influence was memorable-interactions-per-mile [30]. When Knowles [28] reanalyzed the same data, he found that inverse relation between distance and social influence was a measurement artifact, rather than a true indication of how social influence decreases as a function of distance.

Functional Boundaries

As with distance, a team's functional boundaries can be measured in several ways, and we recommend going beyond simply measuring the number of functions represented in a team. In a recent review, Bunderson and Sutcliffe [8] identified three common measures of functional diversity: (1) the extent to which the dominant functional areas differ within the team (members with individual expertise in either engineering, marketing, or manufacturing); (2) the extent to which the functional background mix of each member differs on a team (one member has experience in engineering and marketing, and another on marketing and manufacturing); and (3) the extent to which team members are "assigned" to different functions on a team, regardless of their dominant functional area or various background experiences (a software design engineer is assigned to a software programming team). For all three of these measures, some version of the Blau [6], Shannon [47], or Teachman [48] indices can be used. Another important issue when measuring functional boundaries is to measure them at a level that is meaningful to the participants in the organization. Interviews and pre-research protocols with the participants will help the researcher determine, for example, whether "engineering" is a meaningful functional category, or whether the category needs to be broken down into "systems engineering," "design engineering," and so on.

Temporal Boundaries

When a team has temporal boundaries, the primary implication often is a reduction in the amount of synchronous communication time that is available to the team. O'Leary and Cummings [39] suggest using an "overlap index" to capture this issue, which is based on a measure of shared work hours that represents the available opportunities for synchronous interaction. Such a measure can help control for the presence of multiple time zones or differences in work schedules. Furthermore, the pattern of the temporal boundaries present may be more important than the strict number of temporal boundaries involved. If six out of eight members of the team are working in the same time zone, the remaining members may feel particularly excluded from team communications. Finally, when studying the effects of one boundary, such as geographic distance, researchers should consider including colocated and distributed cases in equivalent synchronous conditions to avoid introducing confounds due to temporal boundaries that may be present.

Identity Boundaries

Existing measures of group identity may be helpful to gauge the strength of team identity [17]. However, these measures do not directly measure the extent to which identities conflict. It may be necessary to measure the number of teams each member belongs to as well as the percentage of work time they devote to the team of interest in your research. As the number of team affiliations increases, so does the opportunity for a worker's identification with those various teams to change over time. With project teams, the level of a member's identification with the team may depend on the member's role and the stage of project development. In situations where members belong to multiple teams, it may be helpful to measure team identity at three points in time to establish a trend line. It may also be useful to consider other proxy measures for identity such as reward systems and reporting structures. Whether members are rewarded as a team or as individuals and whom they report to are often important indirect indicators of their organizational identity.

Organizational Boundaries

In addition to measuring the number of members of the team in different organizations (and the subgrouping patterns of team members across organizations), it may be helpful to consider measures of "organizational distance." Others have suggested that "institutional distance" represents the extent of similarity or dissimilarity between the regulatory, cognitive, and normative institutions of different countries [29, 55]. Such measures could be adapted to reflect the cultural differences between organizations. Moreover, measures of density and centrality may be important in representing patterns of organizational affiliation.

Meta-Observations Across Boundaries

In this paper, we have argued for richer ways of representing boundary variables in field research involving global teams. Now we offer several meta-observations that apply to all of the boundaries we studied and other boundaries as well. First, the effects of many boundaries can change over time, so it is useful to consider repeated data collection at different meaningful points in the life of the team. Second, many of these boundaries can have nonlinear effects. The effects of distance or time differences may be substantial with little distance and time separation, but may not increase too much with further separation [1]. Third, in many cases what may really matter is the pattern or distribution of members along a particular boundary. For instance some preliminary research indicates that groups that are divided into two strong subgroups may experience more conflict than groups that have even greater diversity [44]. Methods for analysis of social networks can be valuable tools to measure and study these patterns [46, 51]. Finally, different measures of boundaries may reveal different effects on dependent variables. Because there still are disagreements about the best way to represent these boundaries, researchers should consider using multiple measures, rather than relying on a single measurement for a particular boundary. The use of multiple measures, or triangulation, may enhance a researcher's ability to evaluate the real effect of that boundary and draw interesting conclusions.

Limitations and Conclusions

We have used empirical evidence from four real organizations to raise awareness of the complex methodological challenges that exist when conducting research across boundaries. Nevertheless, our cross-case study is not without limitations. First, our data and data analysis methods are primarily qualitative. Further quantitative analysis on the interactions and effects of different boundaries would be illuminating. In addition, we have focused only on five specific boundaries, which were salient and relevant across the four case studies on which this research is based. Further studies involving other boundaries (e.g., cultural, political, expertise) are necessary to corroborate and extend our findings.

Despite these limitations, our study has important implications for the design and use of information systems intended to support global collaboration. As others have pointed out [11, 52], information technologies have the potential to help teams bridge boundaries. For example, global collaboration tools could include features to make distant collaborators aware of national and cultural events and issues in other locations (e.g., special holidays, common work hours). Similarly, such collaboration tools could include features to help team members cope with time boundaries (e.g., presence awareness alerting when other team members are online, global time clocks). These tools could help global teams identify their temporal patterns and understand when asynchronous work is more effective (e.g., while coding software) and when synchronous work is preferred (e.g., during software testing and integration).

Our qualitative cross-case analysis highlights the methodological challenges associated with the study of global teams. Specifically, we addressed the potential confounding effects of multiple boundaries, the importance of understanding context to ensure validity, and the role of the individual investigator in determining the salience of specific boundaries. For researchers interested in group processes and virtual organizations, the treatment of these boundaries is a key issue when conducting field studies. Further, these issues are very important to the information systems research community because teams with multiple boundaries are often mediated by information technologies and typically involve virtual, geographically distributed, asynchronous collaboration. Research on global teams has achieved high visibility in the literature; however, little attention has been paid to the methodological implications of studying teams operating in such complex contexts. This paper begins this needed dialogue by identifying the methodological issues as well as some potential design and measurement remedies that can help researchers cope with multiple boundaries.

Acknowledgments: An earlier version of this paper was presented at the Hawaii International Conference on System Sciences, January 7–10, 2002, Big Island, Hawaii. The first author was partly supported by Lucent Technologies and by the National Science Foundation (#IIS-9812123). The second author was supported by the Knowledge and Distributed Intelligence program of the National Science Foundation (#IIS-9872996). The third and fourth authors were supported by the National Science Foundation (#DGE-9354995).

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